

executed via a full day VWAP strategy for large and small cap stocks. The index was computed based on average stock volatility in each category in each month and the corresponding market impact parameters. This index shows how actual costs for this order size change from month to month and also shows the difference in trading costs between large and small cap stocks. Our large cap index was in the 20 – 25 bp range for the majority of the year. The small cap index was in the 40 – 50 bp range over the year but spiked as high as 50 bp in November 2012. The spike in the index in November 2012 was much more dramatic for small cap stocks than for large cap stocks. A short-term cost index such as this helps portfolio managers perform portfolio attribution to understand tracking error and the trading cost environment. It also allows portfolio managers to analyze whether the missed alpha was due to poor stock selection or inferior trading, or whether it was due to unavoidable market conditions.

Some portfolio managers have requested the cost index to be computed based on the I-Star expression without incorporating the strategy or based on an optimized cost index incorporating market impact and timing risk. These strategies are:

I-Star Index (without strategy):

$$I^* = a_1 \cdot \text{Size}^{a_2} \cdot \sigma^{a_3} \quad (12.1)$$

Optimized Cost Index:

$$\text{Min } \mathcal{L} = \text{MI} + \lambda \cdot \text{TR} \quad (12.2)$$

The advantage of using the I-Star calculation (Equation 12.1) is that it is not strategy dependent and so provides managers with a fair comparison of trading costs over time which then can be mapped to any investor strategy. The difficulty is that this value is a proxy for the instantaneous trading cost (ultra-aggressive) and will thus provide cost estimates much higher than will be realized in the market. It works well, however, if used to normalize the change in cost across time (see below).

The advantage of the optimized cost metric (Equation 12.2) is that the cost reflects both market impact cost and timing risk so it also provides investors with a measure of cost and trading difficulty (uncertainty). The difficulty is that this metric will tend to be higher than the expected market impact cost of the trade since it does add the timing risk expression to the total value. Investors further need to specify the risk aversion parameter λ . In this case, investors should use the corresponding MI estimate at the optimized solution.